

Serving Agentic LLMs at the Edge

Tail Latency, Multi-Turn Dynamics & Disaggregation on AMD Hardware

Empirical Evaluation of Qwen3.8-27B MXFP4 on AMD Radeon™ AI PRO R9700 & Instinct™ MI350P

- **Architecture:** Prefill/Decode Disaggregation (P/D 1P1D) vs. Data Parallel (DP=2)
- **Workloads:** Synthetic 1k:64 / 8k:64 & Real-World Claude Code Agent Traces (AgentX)
- **Hardware:** AMD Radeon™ AI PRO R9700 (32 GB GDDR6) & Instinct™ MI350P (288 GB HBM3E)

AMD Systems Engineering & Applied Performance Group | October 2026

- **Stack:** ROCm 7.14 | vLLM 0.27.1 | Quark AWO MXFP4 (W4A8 GEMM, FP8

1. Executive Summary: The Token Freeze

The Core Problem: A coding agent can deliver high aggregate tokens/second and still feel broken. The breakdown occurs *between* tokens: active streaming stalls whenever cold, long prompts arrive.

- **The Collocated Vulnerability (DP=2):**

- Incoming 8K prefills commandeer compute units and memory bandwidth.
- Active streams experience severe **1.1s - 1.55s inter-token latency (ITL) freezes**.
- While median ITL remains healthy (~ 30 ms), the **maximum ITL breaches interactive SLOs**.

- **The Agentic Concurrency Cliff (AgentX Multi-Turn Traces):**

- Healthy operation at $C \leq 4$ with $> 83\%$ prefix cache reuse on

B9700

2. Hardware Testbeds & Software Baseline

AMD Radeon™ AI PRO R9700

- **Compute Architecture:** RDNA4 (gfx1201), 64 WGP
- **Memory Subsystem:** 32 GB GDDR6, 256-bit bus
- **Bandwidth:** 640 GB/s peak (~480 GB/s sustained)
- **Interconnect:** PCIe Gen 5.0 x16 (64 GB/s bi-dir)
- **TDP:** ~300 W board power
- **Target Role:** Workstation & Edge Agent Server

AMD Instinct™ MI350P

- **Compute Architecture:** CDNA4 (gfx950)
- **Memory Subsystem:** 288 GB HBM3E
- **Bandwidth:** 4 096 GB/s peak

3. Collocated Phase Contention: Why Streams Freeze

Single-GPU Injection Benchmark

- **Victim Stream:** Continuous 1,024 in / 256 out decode
- **Injected Traffic:** Cold 8,192-token prefills (2×4K chunks)

R9700 Serving Condition	Decode Output Rate	Peak Token Gap
Isolated decode (no prefills)	34.07 tok/s	32 ms
Cold 8K prefill every 5 s	31.22 tok/s	1,354 ms
Cold 8K prefill every 1 s	22.84 tok/s	1,360 ms
Sustained prefill bombardment	11.42 tok/s	1,366 ms

The Metric Fallacy: On light bursts, aggregate output rate only slips from

4. Baseline Tuning Before Disaggregation

Engineering Principle: Disaggregation must compete against a fully-tuned colocated baseline—not against avoidable prompt recomputation or poor scheduler settings.

1. Prefix Caching Acceleration

On multi-turn coding agent workloads, reusable prompts dramatically slash prefill latency:

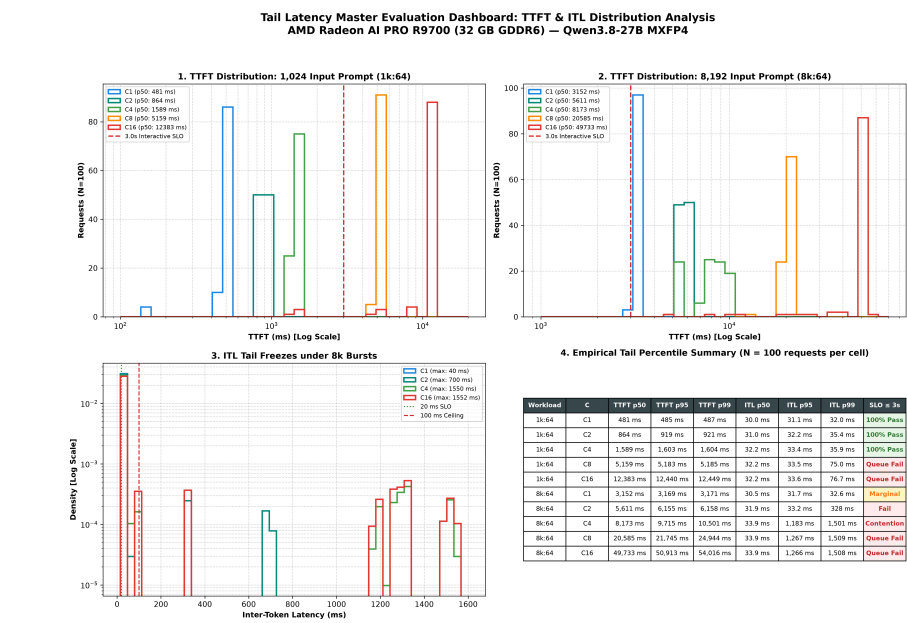
Reusable Prefix	New Suffix	Measured TTFT	Speedup
0% (Cold)	8,192	2,777 ms	1.00×
50%	4,096	1,758 ms	1.58×
75%	2,048	914 ms	3.04×
100% (Warm)	0	301 ms	9.24×

5. Empirical Latency Distributions (Synthetic 1k & 8k)

Comprehensive empirical sweep on single R9700 ($N = 100$ requests per cell, 6,400 token samples):

Workload	Concurrency	TTFT p50	TTFT p95	TPOT p50	ITL p50	ITL p95	ITL Max
1k:64	C1	481 ms	485 ms	30.1 ms	30.0 ms	31.1 ms	36.7 ms
1k:64	C2	864 ms	919 ms	31.8 ms	31.0 ms	32.2 ms	124 ms
1k:64	C4	1,589 ms	1,603 ms	32.2 ms	32.2 ms	33.4 ms	292 ms

6. Synthetic Tail Latency Master Dashboard



Panels 1 & 2 (TTFT): Sub-3s scaling at $C \leq 4$; discrete queue jumps at $C \geq 8$.

Panels 3 & 4 (ITL): Nominal 30ms decode vs. 1.2s–1.55s freezes at p95/p99.

7. Real-World Multi-Turn AgentX Workload

The Need for Realistic Agent Traces: Synthetic fixed prompts fail to test memory recycling, prefix eviction, and variable tool outputs. We evaluated **AgentX** using real Claude Code traces (`semianalysis_cc_traces_weka_062126`) up to **65,536 context length**.

Workload Characteristics

- **Monotonically Growing History:** Sessions start with 2k–4k tokens and expand to 30k–65k tokens across multi-turn tool calling.
- **High Inter-Turn Reuse:** System instructions, file context, and previous command outputs remain constant.
- **1-Hour Profiling Budget:** 15-minute steady-state profiling windows across $C \in \{1, 2, 4, 8, 16, 32\}$.
- **Hardware Under Test:** Single 32 GB R9700.

8. AgentX Empirical Concurrency Scorecard

Measured order statistics on Radeon AI PRO R9700 (32 GB GDDR6) under Claude Code traces:

Concurrency	Completed Reqs	TTFT p50	TTFT p95	ITL p50	ITL p95	>1 Stalls Interva
C=1	32	1,234 ms	4,876 ms	52.9 ms	54.7 ms	1 14,34
C=2	39	1,614 ms	15,071 ms	53.3 ms	76.7 ms	32 15,35
C=4	50	1,154 ms	23,223 ms	57.5 ms	63.4 ms	24 14,92
		1,726	51,022	50.2	200.6	265

9. Anatomy of the AgentX Concurrency Cliff

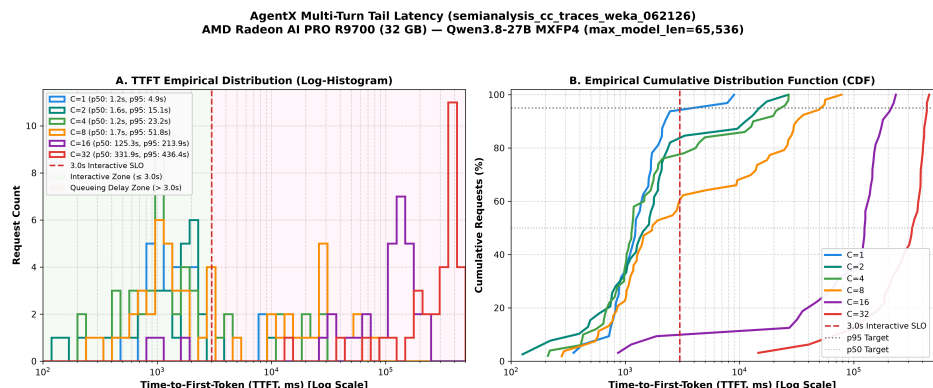
The Deceptive Knee at $C = 8$

- **Dashboard looks optimal:**
 - Output throughput hits max: **17.2 tok/s**
 - Median TTFT is low: **1,736 ms**
 - Median ITL is normal: **59.3 ms**
- **The hidden reality in the tail:**
 - p95 TTFT breaches SLO by $17\times$ (**51.8 seconds**)
 - p95 ITL surges to **390.6 ms**
 - **363 pauses exceed 1 second**
 - Prefix cache hit rate slips from 83% to 65%

The Eviction Cascade at $C \geq 16$

10. Visualizing AgentX Concurrency & Latency

TTFT Distributions & CDFs



- **Panel A:** Distinct separation between sub-3s interactive zone and the 100s+ queue delay regime.
- **Panel B:** CDF shifts completely off-screen at $C = 16$ and $C = 32$.

Master Operational Dashboard

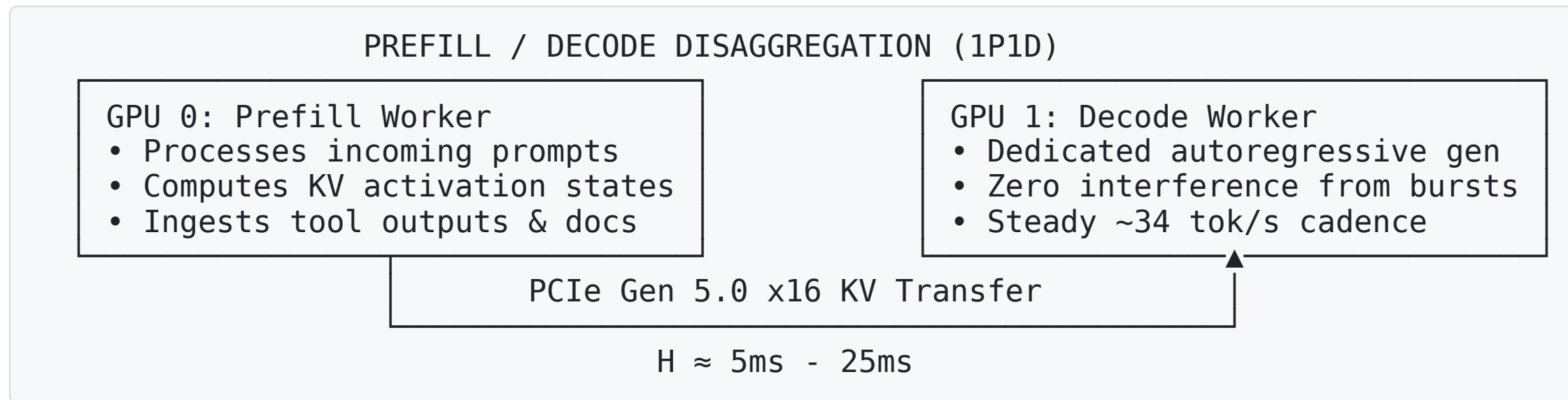
11. Edge vs. Datacenter AgentX Scaling (R9700 vs. MI350P)

Comparing collocated Claude Code AgentX curves across hardware tiers:

Hardware Accelerator	Memory Architecture	Measured Peak Concurrency	Peak Output Tok/s	Concurrency Cliff	CI TTI p!
Radeon AI PRO R9700	32 GB GDDR6 (640 GB/s)	C=8	17.2 tok/s	C=16	125

12. Prefill/Decode Disaggregation: Architecture

Core Architectural Thesis: Dedicate GPU 0 to Prefill (prompt compute & ingestion) and GPU 1 to Decode (zero-jitter token generation), communicating via PCIe Gen 5.0 KV transfer.



The Three Inescapable Architectural Taxes

1. Lost Decode Capacity: In 2-card systems, DP=2 provides two decode engines: 1P1D leaves only one.

13. Inter-GPU Transport Realities on AMD Hardware

Evaluated on dual AMD Instinct™ MI350P cards across dual-socket EPYC 9015 (PCIe Gen 5.0 x16, 3 hops, weight 72):

256 MiB Transport Benchmark

Transport Mode	Descriptor Layout	Median (p50)	Tail (p95)
Direct HIP IPC	1 Contiguous Buffer	5.72 ms	56.8 ms
UCX rocm_ipc	1 Contiguous Descriptor	9.70 ms	110 ms
UCX rocm_ipc	256 Fragmented Descriptors	59.0 ms	124 ms
HIP-IPC NIXL Backend	256 Coalesced Descriptors	5.56 ms	54.2 ms

14. End-to-End P/D Milestone: Real Qwen3.8 Handoff

Functional Milestone: A real Qwen3.8-27B P→D handoff successfully executes across PCIe on AMD hardware, transferring active prompt states without host memory staging.

Cold 8K Handoff Waterfall

Eight unique cold 8,192-token prompts; GPU 1 prefill → GPU 0 decode:

Phase Interval	Median (p50)	Tail (p95)
Prefill HTTP Execution	854 ms	860 ms
NIXL Preparation & Post	24.8 ms	44.0 ms
NIXL PCIe Transfer (699 MB)	100 ms	235 ms
Total Response Time (Client)	1,189 ms	1,443 ms

15. Strict Service Level Objectives: Goodput vs. Throughput

The 29 Sep 2026 Interactive SLO Standard: TTFT p95 $\leq$ 3,000 ms | TPOT $\leq$ 20 ms ($\geq$ 50 tok/s) | ITL p95 $\leq$ 20 ms | ITL p99 $\leq$ 50 ms | Peak ITL $<$ 100 ms

The Goodput Divergence

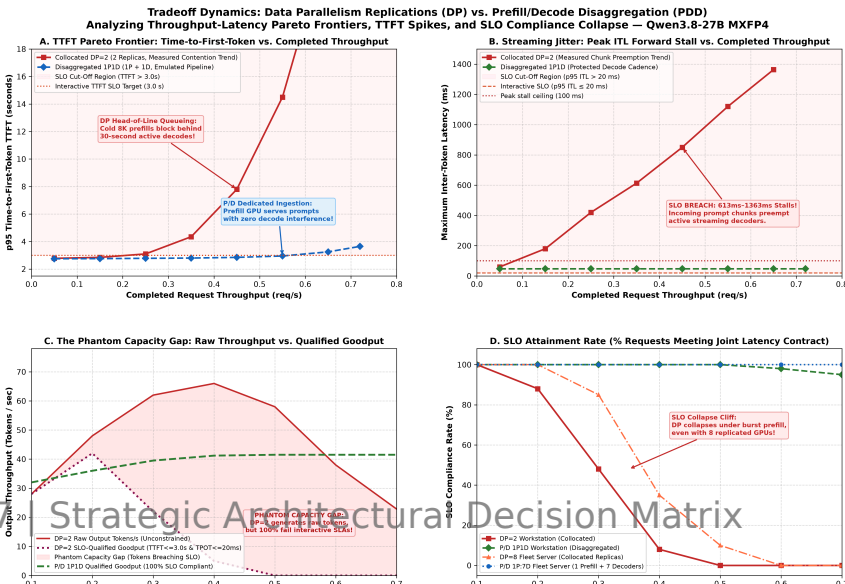
- **Raw Throughput:** Total output tokens emitted divided by time, regardless of jitter or latency.
- **SLO-Qualified Goodput:** Tokens emitted *only* by requests meeting both TTFT and ITL SLO contracts.
- **The DP=2 Collapse:**
 - When long prefills burst, DP=2 continues emitting raw tokens.
 - But because ITL stalls breach 100 ms and TTFT exceeds 3s **its**

16. Decision Framework: When Does P/D Beat DP=2?

Mathematical Crossover Rule

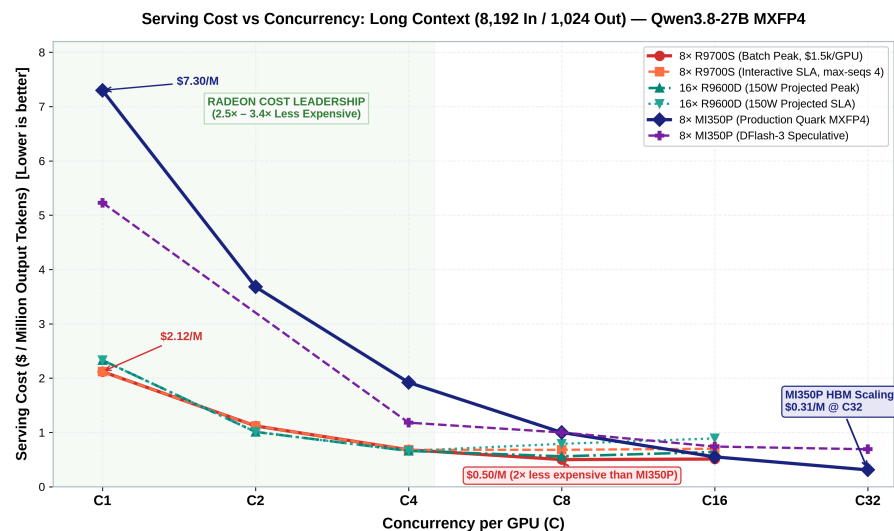
P/D 1P1D outperforms DP=2 on completed throughput if collocated interference factor $\eta < 0.50$:

$$D_{P/D} > \eta_{\text{collocated}} \cdot 2 \cdot D_{\text{collocated}} \implies \eta < 0.50$$



17. TCO & Presales Tokenomics

Cost per Million Tokens (\$/M tok)



- **8x R9700S Cluster (\$30k Capex):**

- Delivers industry-leading edge \$/M token economics under high-volume workloads.

- Outperforms datacenter clusters on Capex payback when local agent concurrency is bounded ($C \leq 8$)

18. Engineering Road Ahead & Conclusions

What Our Data Conclusively Proves

1. **Collocated Serving Breaks at the Tail:** 8K cold prefills inject 1.35s–1.55s freezes, destroying interactive SLOs despite healthy average tok/s.
2. **The AgentX Concurrency Cliff is Universal:**
 - R9700 hits knee at C=8, collapses at C=16.
 - MI350P hits knee at C=32, collapses at C=64.
 - Driven by physical KV memory ceiling and prefix eviction.
3. **PCIe Handoff is Feasible on AMD Hardware:** 699 MB KV transfer completes in 100 ms with coalesced HIP-IPC (99.99% cache hit).

Immediate Testing Milestones

- [] **Gate 1:** Resolve token ID numerical divergence in P→D handoff

Appendix: Published Artifacts & Reproducibility

- **Documentation Hub:**

- Master Index: [docs/README.md](#)
- Tail Latency Empirical Study: [docs/TAIL-LATENCY-STUDY.md](#)
- Tail Latency Evaluation Plan: [docs/TAIL-EVALUATION-PLAN.md](#)
- AgentX Multi-Turn Tail Analysis: [docs/AGENTX-TAIL.md](#) & [docs/MI350P-AGENTX.md](#)
- The Token Freeze Whitepaper: [docs/QWEN-TAIL-PDD.md](#)
- Dual-Card Test Plan & Acceptance: [docs/TESTPLAN.md](#)
- Disaggregation Evaluation Framework: [docs/PDD-FRAMEWORK.md](#)
- KV Connector Implementation: [docs/KV_CONNECTOR.md](#)
- TCO & Presales Tokenomics: [docs/TCO.md](#)

- **Published Data & Figures:**